

We use different metrics to assess both the performance and accuracy of each method on a variety of types of sequencing samples. These experiments are designed to cover a variety of different use-cases for an aligner, spanning the gamut from situations where most alignments are expected to be unique (DNA-seq), to situations where each fragment is expected to align to many loci with similar quality (RNA-seq and metagenomic sequencing), and spanning the range of index sizes from small transcriptomes to large collections of genomes.

PuffAligner's underlying index, Pufferfish supports two index variants, 'dense', and 'sparse' (Dense is the default index and sparse is activated using the '-s' option in the index build step). The 'sparse' version trades off lookup time for index size, and provides a smaller index in which k -mer lookups take longer. However, in the alignment process described here, since the steps after the initial lookup are the main bottleneck in alignment time (e.g. compacting uniMEMs into MEMs, chaining MEMs and finally performing between-MEM alignment), we recommend the sparse Pufferfish index to be the choice. This is the index version we have used for all experiments in this manuscript.

First, we show PuffAligner exhibits similar accuracy for aligning DNA-seq reads to Bowtie2, but it is considerably faster. In the case of experimental reads, since the true origin of the read is unknown, we use measures such as mapping rate and concordance of alignments to compare the methods. Furthermore, in [Supplementary Section S7](#) we evaluate the accuracy of aligners by aligning simulated DNA-seq reads that include variation (single-nucleotide variants and small indels with respect to the reference). For aligning RNA-seq reads, we compare the impact of alignments produced by each aligner on downstream analysis such as abundance estimation. Finally, we show PuffAligner is very efficient for aligning metagenomic samples where there is a high degree of shared sequence among the reference genomes being indexed. We also illustrate that using alignments produced by PuffAligner yields the highest accuracy for abundance estimation of metagenomic samples.

3.1 Alignment of whole genome sequencing reads

First, we evaluate the performance of PuffAligner with a whole genome sequencing (WGS) sample from the 1000 Genomes project ([Consortium et al., 2015](#)). We downloaded the ERR013103 reads from sample HG00190, which is a low-coverage sample from a Finnish male, sequenced in Finland (<https://www.internationalgenome.org/data-portal/sample/HG00190>). There are 18 297 585 paired-end reads, each of length 108 nucleotides in this sample. Using fastp ([Chen et al., 2018](#)), we remove low quality ends and adapter sequences from these reads. After trimming, there are 15 404 412 reads remaining in the sample. Indices for each of the tools are built over all DNA chromosomes of the human genome [GRCh38 ([Schneider et al., 2017](#))] which is obtained from gencode release v33 ([Frankish et al., 2019](#)) (https://www.encodegenes.org/human/release_33.html). All the tool, if possible, are run in the concordant-mode in this experiment.

The alignment rate, run-time memory usage and running time for all the aligners are presented in [Table 1](#). The reason that deBGA has the highest mapping rate in [Table 1](#) compared to other tools is that it is local alignments for the reads that are not alignable end-to-end under the scoring parameters for the other tools. Bowtie2 and PuffAligner are both able to find end-to-end alignments for about

~95% of the reads. STAR and PuffAligner are the fastest tools, with STAR being somewhat faster than PuffAligner. On the other hand, PuffAligner is able to align more reads than STAR, while requiring less than half as much memory. The memory usage of Bowtie2 is the smallest, since Bowtie2's index does not contain a hash table. However, this comes at the cost of having the longest running time compared to other methods. Overall, PuffAligner benefits from the fast query of hash-based indices while its run-time memory usage, which is mostly dominated by the size of the index, is significantly smaller than other hash-based aligners. Although deBGA's index is based on the de Bruijn graphs, similar to the Pufferfish index, the particular encoding for it is not as space-efficient as that of Pufferfish.

To look more closely how the mappings between the tools differ, we investigate the agreement of the reads which are mapped by Bowtie2, STAR and PuffAligner. We are only comparing these three methods which perform end-to-end alignment in this plot, since outliers from the local alignments computed by deBGA would otherwise dominate the plot. Majority of the reads (~14.2M reads) are mapped by all three aligners. The next largest set (~400K reads) represents the reads which are only mapped by Bowtie2 and PuffAligner. All the other sets are much smaller compared to the first two sets. This fact illustrates that the highest agreement in the aligners is between Bowtie2 and PuffAligner. We have also visualized the results in an upset plot in [Supplementary Figure S1](#) using the UpsetR library ([Conway et al., 2017](#)).

Exploring a series of individual reads from the smaller sets in the upset plot, suggests that some of these differences happen as a result of small differences in the scoring configuration, while some result from different search heuristics adopted by the different tools. [Supplementary Figure S2](#) shows the coherence between the alignments reported by the tools by also including the exact location to which the reads are aligned in the reference.

3.2 Transcript abundance estimation from RN-seq reads

Mapping sequencing reads to target transcriptomes is the initial step in many pipelines for reference-based transcript abundance estimation. While lightweight mapping approaches ([Bray et al., 2016](#); [Patro et al., 2017](#)) greatly speed-up abundance estimation by, in part, eliding the computation of full alignment between reads and transcripts, there is evidence that alignments still yield the most accurate abundance estimates by providing increased sensitivity and avoiding spurious mappings ([Sarkar et al., 2018](#); [Srivastava et al., 2020](#); [Vuong et al., 2018](#)). Thus, the continued development of efficient methods for producing accurate transcriptome alignments of RNA-seq reads remains a topic of interest.

In [Supplementary Section S6](#), we compare the effect of alignments produced by each tool on the accuracy of RNA-seq abundance estimation. We find that RNA-seq quantification based on alignments produced by PuffAligner, STAR and Bowtie2 reaches the same level of accuracy. However, PuffAligner is the fastest aligner, being at least $1.5\times$ faster than STAR and $2.5\times$ faster than Bowtie2 while the memory usage by PuffAligner is only $2\times$ larger than memory used by Bowtie2 and $3.4\times$ smaller than STAR (full details in [Section 6](#)).

3.3 Alignment to a collection of microorganisms—simulated short reads

One main property of metagenomic samples is that they contain reads from a variety of genomes. Some of these genomes are highly similar and some are not even assembled yet—and hence unknown. To demonstrate the performance and accuracy of PuffAligner for metagenomic samples, we design two different experiments. The first which we call 'Single-strain' experiment is designed to specifically evaluate issues related to the similarity challenge and the second, the 'bulk' experiment, evaluates aligners in the presence of a high variety of species in the sample in addition to the high similarity of references. We discuss the results of the 'bulk' experiment which is a more comprehensive one, in the following paragraphs. The 'single-strain' experiment is described in [Supplementary Section S8](#).

Table 1. The performance of different tools for aligning experimental DNA-seq reads

Aligner	Mapping-rate (%)	Time (mm:ss)	Memory (GB)
PuffAligner	95.58	6:14	13.09
deBGA	99.75	10:46	41.04
STAR	93.88	4:29	30.36
Bowtie2	95.44	16:15	3.50

Note: The time reports are benchmarked after warming up the system cache so that the influence of index loading time is mitigated.

We construct the indices of PuffAligner, Bowtie2, STAR and deBGA on a random set of 4000 complete bacterial genomes downloaded from the NCBI microbial database. [Supplementary Table S6](#) shows the time and memory required for constructing each of the indices, in addition to the size of the final index on disk. Overall, PuffAligner and Bowtie2 show a similar trend in time and memory requirements, while STAR and deBGA require an order of magnitude more memory. In terms of the final index size, Bowtie2 has the smallest index, PuffAligner has the second-smallest, and STAR has the largest.

We select three Illumina WGS samples with accession IDs SRR10948222 ([Fisher, 2020](#)), SRR11283975 and SRR11496426, the details of which are explained in [Supplementary Table S4](#) and simulate ~50M reads for each through the process explained in [Supplementary Section 9.1](#).

The assessment of ‘accuracy’ directly from the aligned reads is not a trivial task. Due to the high rate of multi-mapping in these simulated samples, and due to the fact that multiple references can produce alignments of the same quality as the ‘true’ origin of the read, we calculate the accuracy by comparing the true and estimated abundances using a quantification tool (in this case, Salmon) rather than by comparing the read alignments directly. In [Table 2](#) the accuracy metrics are calculated over the abundance estimations obtained using the alignments produced by running the aligners in the different modes specified. The list of metrics for metagenomic expression evaluations are *Spearman Correlation*, *Mean Absolute Relative Difference (MARD)*, *Mean Absolute Error (MAE)* and *Mean Squared Log Error (MSLE)*. The metrics have been chosen to be similar to previous work such as in Bracken ([Lu et al., 2017](#)) and Karp ([Reppell and Novembre, 2018](#)). The definition of each of these metrics is provided in [Supplementary Equation S2](#) in [Supplementary Material](#).

This experiment leads to three main observations. First, regardless of the alignment mode, quantifications derived from the deBGA alignments seem to lead to systematic underestimation of abundance. However, PuffAligner, STAR and Bowtie2, show very similar behavior with respect to accuracy. STAR is the best in primary mode as well as when allowing 20 alignments, closely followed by PuffAligner. When allowing up to 200 alignments per read, Bowtie2 tends to yield the most accurate abundances, again with PuffAligner being the close runner-up. These results demonstrate that PuffAligner is a reliable alignment tool showing a stable pattern of

Table 2. Accuracy of abundance estimation with Salmon using alignments reported by each aligner for the mock metagenomic sample simulated from SRR10948222

Alignment mode	Tool	Spearman	MARD	MAE	MSLE
Primary	PuffAligner	0.69	0.028	1.39	0.08
	Bowtie2	0.58	0.053	2.91	0.15
	STAR	0.727	0.023	1.493	0.05
	deBGA	0.28	0.616	656.08	6.53
Up to 20	PuffAligner	0.9	0.006	0.40	0.006
	Bowtie2	0.85	0.01	0.22	0.012
	STAR	0.929	0.004	0.303	0.002
	deBGA	0.28	0.573	637.60	5.65
Up to 200	PuffAligner	0.97	0.002	0.36	0.001
	Bowtie2	0.99	0.001	0.19	0.00
	STAR	0.929	0.004	0.299	0.002
	deBGA	0.28	0.571	637.83	5.55
Best strata	PuffAligner	0.97	0.002	0.36	0.001
	STAR	0.929	0.004	0.3	0.002

Note: All aligners are run in three main modes; allowing only one best alignment with ties broken randomly (Primary), up to 20 alignments reported per read and up to 200 alignments reported per read. PuffAligner and STAR support a fourth mode that allows reporting all equally best alignments (bestStrata). This option improves the performance while maintaining the accuracy of the results. Best result in each metric is highlighted in bold.

being comparable to the best aligner under all the scenarios tested. That is, the good performance of PuffAligner is robust across a variety of different parameter settings.

Moreover, due to the nature of the metagenomic data—the high degree of ambiguity and multi-mapping—we expect to see improvement in the accuracy metrics as more alignments are reported per read, as this leads to a higher recall. While STAR’s accuracy changes only slightly from 20 alignments to 200 alignments (only improving MAE) the results for PuffAligner and Bowtie2 improve considerably when allowing more alignments per read. However, this higher accuracy comes in the cost of alignment time for Bowtie2 as shown in [Figure 3](#). The difference becomes especially evident when allowing up to 200 alignments per read, where PuffAligner is 4 times faster than Bowtie2. In addition, in experimental data, many of the alignments reported do not necessarily have high quality, and only appear in the output as one of the 200 alignments for the read. In fact, we note the similar accuracy achieved by PuffAligner in *bestStrata* mode compared to when we allow up to 200 alignments per read. This observation is also consistent across the other two simulated samples in [Supplementary Table S5](#), in those cases with PuffAligner being the most accurate aligner in different modes for both samples.

Overall, these results along with other similar experiments in [Supplementary Section S8](#) and [Supplementary Table S5](#) indicate that PuffAligner is a sensitive and fast aligner. Specifically PuffAligner exhibits similar accuracy (and is sometimes more accurate) as well-known aligners like Bowtie2 and STAR. On these data, it exhibits memory requirements close to those of the memory-frugal Bowtie2, while being much faster. [Figure 3](#) shows that PuffAligner has the lowest running time, even when the number of allowed alignments per read increases. In [Supplementary Section S11](#), we explain the reasons and advantages behind using PuffAligner as opposed to a light-weight pipeline like Kraken2 + Bracken for metagenomic analyses with examples from real metagenomic experiments.

3.4 Scalability

[Figure 2](#) and [Supplementary Figure S7](#) represent how the construction memory and index size of each tool scales over different types of sequences. The trend shows the effect of database size as well as redundancy and sequence similarity on the scalability of each of the tools. Tools such as PuffAligner and deBGA, which build a de Bruijn graph-based index on the input sequence, specifically compress similar sequences into unitigs and therefore scale well for databases with high redundancy such as collections of related microbial genomes. As the figure shows, the size of the PuffAligner index increases relatively less compared to that of Bowtie2 and STAR as we add more (specifically, repetitive) sequences so that, although the PuffAligner index is almost three times larger than the Bowtie2 index for the human genome, PuffAligner’s index is similar in size to that of Bowtie2 over the references for the metagenomic experiments. It is worth mentioning that Bowtie2 requires a switch from a 32-bit index to a 64-bit index as the total count of the input bases increases,

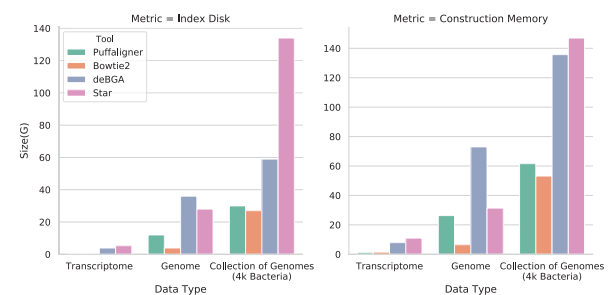


Fig. 2. Time performance for aligning a mock experiment simulated from bulk read sample SRR10948222. The dashed area shows fraction of the time spent purely on aligning reads and the rest is the time required for index loading. PuffAligner is the fastest tool, yet most of its time is still dedicated to loading the index. The alignment for Bowtie2 increases when asking for more alignments per read while the other tools show a constant alignment time scaling over number of reads

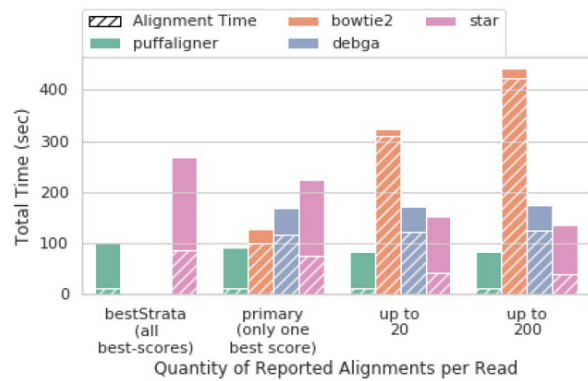


Fig. 3. Scalability of different tools over the final index disk and c onstruction memory for three different datasets, human transcriptome (encode version 33), human genome (GRCh38 primary assembly) and collection of genomes (4000 random bacterial complete genomes)

which is another reason why the size is growing super-linearly. To better evaluate how the different indices grow as a function of the size of the reference database and similarity of the sequences that constitute the database, we designed a small-scale experiment where we compare the scalability of Bowtie2 and PuffAligner when increasing the size of the database under two different scenarios. In one case, we index a collection of increasing size of highly similar genomes, and in the other we index a collection of increasing size of genomes from distinct species (with much less sequence similarity). In [Supplementary Section 10](#), we demonstrate the scalability advantage of PuffAligner for indexing highly redundant and similar reference collections.

4 Discussion and conclusion

In this article, we introduce PuffAligner, an aligner suitable for the contiguous alignment of short-read sequencing data. We demonstrate its use in aligning DNA-seq reads to the genome of a single species, aligning RNA-seq reads to the transcriptome, and aligning DNA-seq reads from metagenomic samples to a large collection of references. It is built on top of the Pufferfish index, which constructs a colored compacted de Bruijn graph using the input reference sequences. PuffAligner begins read alignment by collecting unique maximal exact matches, querying k -mers from the read in the Pufferfish index. The aligner then chains together the collected uniMEMs using a dynamic programming approach, choosing the chains with the highest coverage as potential alignment positions for the reads. Finally, PuffAligner is able to efficiently compute alignment, exploiting information from long matches in the chains and making use of an alignment cache to avoid redundant work.

We compared the accuracy and efficiency of PuffAligner against two widely used alignment tools, Bowtie2 and STAR, that perform unspliced and (optionally) spliced alignments of reads, respectively. We also compare the results against deBGA, an aligner that also utilizes an index built over the compacted de Bruijn graph.

We analyze the performance of these tools on both simulated and experimental DNA and RNA sequencing datasets. The accuracy of PuffAligner is comparable to Bowtie2, which exhibits very high alignment. PuffAligner generally performs better than STAR and deBGA (though, unlike STAR, none of these other tools currently support spliced read alignment). In terms of speed and memory, PuffAligner reaches a tradeoff between the relatively high memory usage of STAR and deBGA and the slower speed of Bowtie2. Hence, while the memory requirement of PuffAligner is more than that of Bowtie2, the speed gain is significant. In the tests performed in this manuscript, PuffAligner is almost always the fastest tool (with the exception being that STAR is faster when aligning unspliced DNA-seq reads to a single human genome).

An additional advantage of the Pufferfish index used in PuffAligner is that it can scale well to a collection of genomes, transcriptomes, or both, especially when the reference sequences being indexed exhibit a high degree of sequence similarity. This feature is already utilized in a specific pipeline for RNA-seq quantification that makes use of a joint index over the genome and transcriptome (Srivastava *et al.*, 2020). The analysis shows that specificity of alignments in such a case can be improved by filtering from quantification reads that are better aligned to some genomic locus that is not present in the transcriptome.

Furthermore, the nature of the Pufferfish index, that explicitly factorizes out highly repetitive sequence, coupled with the fast (and repetition-aware) alignment procedure of PuffAligner makes it a particularly useful for indexing and aligning to a highly similar collection of sequences. This potentially makes it a good match for metagenomic analyses.

We have provided a proof of concept for such a PuffAligner-based metagenomic analysis pipeline, and plan to build a more sophisticated and fully featured metagenomic analysis framework around PuffAligner in the future.

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